INTERNATIONAL UNIVERSITY OF APPLIED SCIENCES

Advanced Statistics Workbook

University of Applied Science - Online

Study-branch: Data Science

Advanced Statistics Workbook

Abel Yann Djabou Nghongnang

Matrikelnummer: IU14096047

Straße 10

realized in....

Advisor: Paul Libbrecht

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III Abbreviations

AFL American Fuzzy Lop
API Application Programming Interface
BIOS Basic Input/Output System
Brick Binary Run-time Integer Based Vulnerability Checker
CaaS Container as a Service
CAB Change Advisory Board
CE Community Edition
CI Continuous Integration
CLI Command Line Interface
CNCF Cloud Native Computing Foundation
CRED C Range Error Detector
Dev Development, the development team

1 Latex

1.1 Tools

MiKTeX: https://miktex.org/download TeXLive: http://tug.org/texlive/ (or alternative LaTeX-systems).

A good editor is essential. Sometimes combined editors and compilers (e.g. TeXShop) can be really productive. Make sure you learn the use of synchronized navigation then.

A vector graphic is one where strokes remain strokes even at the highest resolutions: e.g. the Figure 1 or the Logo on the Titelblatt (notice: you can click from here to there). Many tools generate vector-graphics for plots from any data-set. E.g. Plotly (with the use of the Browser-Print), MatPlotLib or even OpenOffice, LibreOffice or MS-Excel.

1.2 Literature References

Here is an example of a reference with a page-number: Dueck (2016, p. 6)

1.3 Pictures

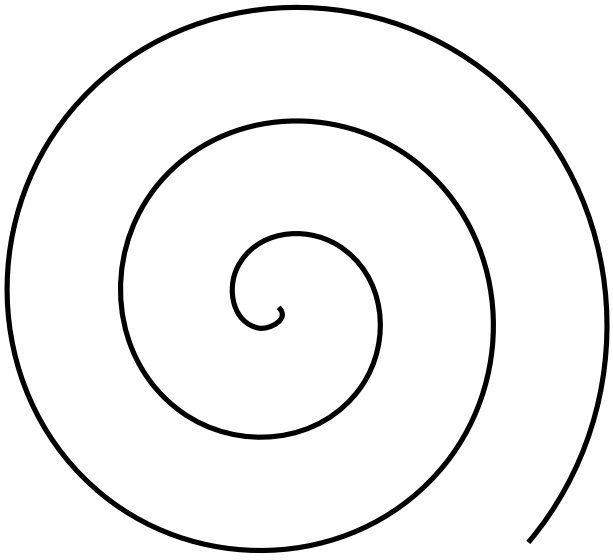


Figure 1: A spiral... smooth vector-based with a clean parametrisation!
Nothing to do with Gage (2018)

1.4 Tables

“ Industrial era ”	“Jobs ”	“ Wanted: Upgrade”’
Parts exchanger	Fitter	mecatronics specialist
eShop	reseller	“ Client-suggester”
“ Coding-guru”’	Softwaredesign	Whole-life designer
JA! Gut & Günstig	brand-names	“ Life-Style Feeling”
Internetbanking	Bank clerk	Customer adviser
Robots	Specialist	Machine supervisor
Bush	Gardener	Nature-sculptor
Painting	Painter	Interior Design

Table 1: Downgrade and Upgrade of job denominations
Dueck (2016)

1.5 Listes

- one
- twoi
- threei

1. first
2. second
3. third

1.6 Formulæ

A formula can be written inline, e.g. as $\frac{d}{dx}\text{arctg}(x) = \frac{1}{1+x^2}$ or, in centered math:

$$\frac{d}{dx}\text{arctg}(x) = \frac{1}{1+x^2} \tag{1.1}$$

Notice that formulæ that are centered start bigger (technically, they start in `\displaystyle`) than they start inline (technically, they start in `\textstyle` all subsequents reductions, e.g. an exponent, goes to `\scriptstyle` then `\scriptscriptstyle`). Indeed a best effort is made so that inline formulæ do not change the line height which would bother the eye of a reader.

Formulæ can be given a number and a label. Numbering happens automatically with `\begin{equation}` and `\end{equation}` and can be avoided if enclosing the formula between `\[` and `\]`. If using the `\label` macro inside, you can refer automatically to this equation using `\ref{label}`. E.g. Thanks to equation 1.1 one dare say that:

$$\int_0^t \frac{1}{1+x^2}dx = \arctan(t) \tag{1.2}$$

1.7 Tools and Code

Many users of this template will want to include some code.

The simplest way to do so is to use the `\verb` macro which is followed by a sign, then some code, then the sign again to close. This is the inline version which works as in:

```
As we could calculate with \textcite{Wolfram_Alpha} using
\verb_integrate 1 / pi e ^ (t/pi) from zero to infinity_.
```

which yields:
As we could calculate with Wolfram Inc. (2021) using `integrate 1 / pi e ^ (t/pi)` from zero to infinity.
The multiline version of this is called `\begin{verbatim}` and finishes with `\end{verbatim}`.

1.8 Citation examples

Monography (Liebel, 2019, p. 22)
Collection (Müller & Mustermann, 2019)
Article (Schülers, 2021)

Bibliography

Dueck, G. (2016). Zwischen industralisierendem Downgrade und notwendigem Upgrade/Empowerment [auf dem AKAC-Kongress].

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Wolfram Inc. (2021). Wolfram Alpha [Accessed: 10.06.2021]. https://wolframalpha.com

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Eidesstattliche Erklärung

I hereby certify...

.....	.....
Place, date	Signature